

AI Search Strategy Questions (With Answers)

Generated from Knowledge Graph Analysis

This document contains problem instances for various AI search problems and questions about the most appropriate solving strategies. Each problem includes instance visualizations and detailed answers based on knowledge graph analysis.

1. N-Queens

Instance 1:

Board Size: 4×4, Pre-placed Queens: 2

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Question: For the N-Queens problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 2:

Board Size: 4×4, Pre-placed Queens: 0

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Question: For the N-Queens problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 3:

Board Size: 5×5, Pre-placed Queens: 0

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Question: For the N-Queens problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

2. Tower of Hanoi

Instance 1:

Number of Disks: 3

Initial Configuration:

Peg A: [3]

Peg B: [2, 1]

Peg C: []

Goal: All disks on Peg C

Question: For the Tower of Hanoi problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 2:

Number of Disks: 4

Initial Configuration:

Peg A: [4, 3, 2, 1]

Peg B: []

Peg C: []

Goal: All disks on Peg C

Question: For the Tower of Hanoi problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 3:

Number of Disks: 4

Initial Configuration:

Peg A: [4, 3, 2, 1]

Peg B: []

Peg C: []

Goal: All disks on Peg C

Question: For the Tower of Hanoi problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

3. Graph Coloring

Instance 1:

Vertices: 5, **Colors Available:** 4, **Edges:** 8

Graph Edges:

(0, 1), (0, 2), (0, 3), (0, 4), (1, 2), (1, 3), (1, 4), (3, 4)

Question: For the Graph Coloring problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 2:

Vertices: 7, **Colors Available:** 3, **Edges:** 14

Graph Edges:

(0, 1), (0, 2), (0, 3), (0, 4), (1, 2), (1, 3), (1, 4), (1, 5), (2, 4), (2, 5), (2, 6), (3, 5), (4, 6), (5, 6)

Question: For the Graph Coloring problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 3:

Vertices: 7, **Colors Available:** 4, **Edges:** 12

Graph Edges:

(0, 6), (1, 2), (1, 3), (1, 4), (1, 5), (2, 3), (2, 4), (2, 6), (3, 5), (3, 6), (4, 5), (5, 6)

Question: For the Graph Coloring problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

4. Knight's Tour

Instance 1:

Board Size: 5×5, **Starting Position:** (0, 3), **Visited Squares:** 5

.	.	.	1	.
.	.	.	.	5
.	.	4	.	2
.	.	.	.	.

.	.	.	3	.
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Question: For the Knight's Tour problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 2:

Board Size: 6×6, Starting Position: (3, 2), Visited Squares: 5

.	.	.	.	.	3
.	.	.	4	.	.
.	.	.	.	2	5
.	.	1	.	.	.
.	.	.	.	.	.
.	.	.	.	.	.

Question: For the Knight's Tour problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 3:

Board Size: 6×6, Starting Position: (2, 5), Visited Squares: 8

.	.	.	.	.	.
.	.	.	.	3	.
.	.	.	5	.	1
.	8	.	2	.	4
.	.	6	.	.	.
7	.	.	.	.	.

Question: For the Knight's Tour problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

5. 8-Puzzle

Instance 1:

Initial State (Goal: 1-2-3-4-5-6-7-8-blank)

7	4	1
	2	3
5	8	6

Question: For the 8-Puzzle problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 2:

Initial State (Goal: 1-2-3-4-5-6-7-8-blank)

4	1	3
2	6	8
7	5	

Question: For the 8-Puzzle problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

Instance 3:

Initial State (Goal: 1-2-3-4-5-6-7-8-blank)

1	2	3
7		8
6	4	5

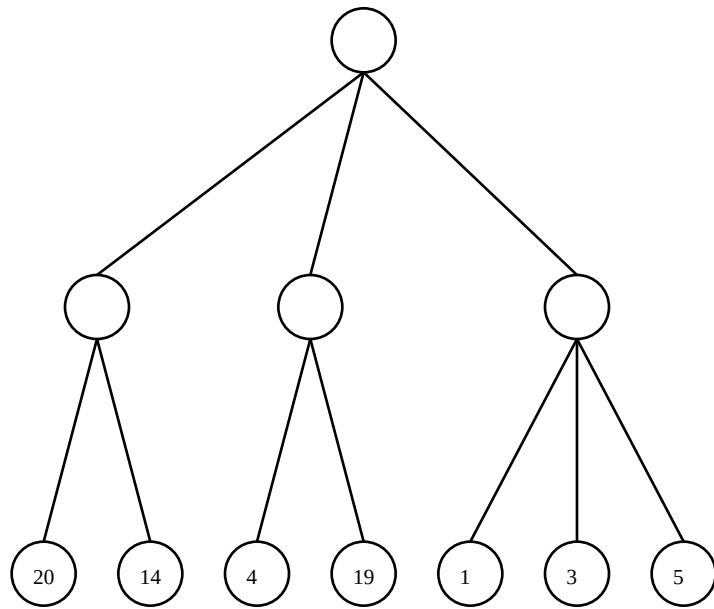
Question: For the 8-Puzzle problem and the given instance, which is the most appropriate solving strategy among those mentioned in the course (BFS, DFS, UCS, A*, GBFS, IDA*, Hill Climbing, Simulated Annealing)?

Answer:

No specific recommendations available.

6. MinMax

Instance 1:



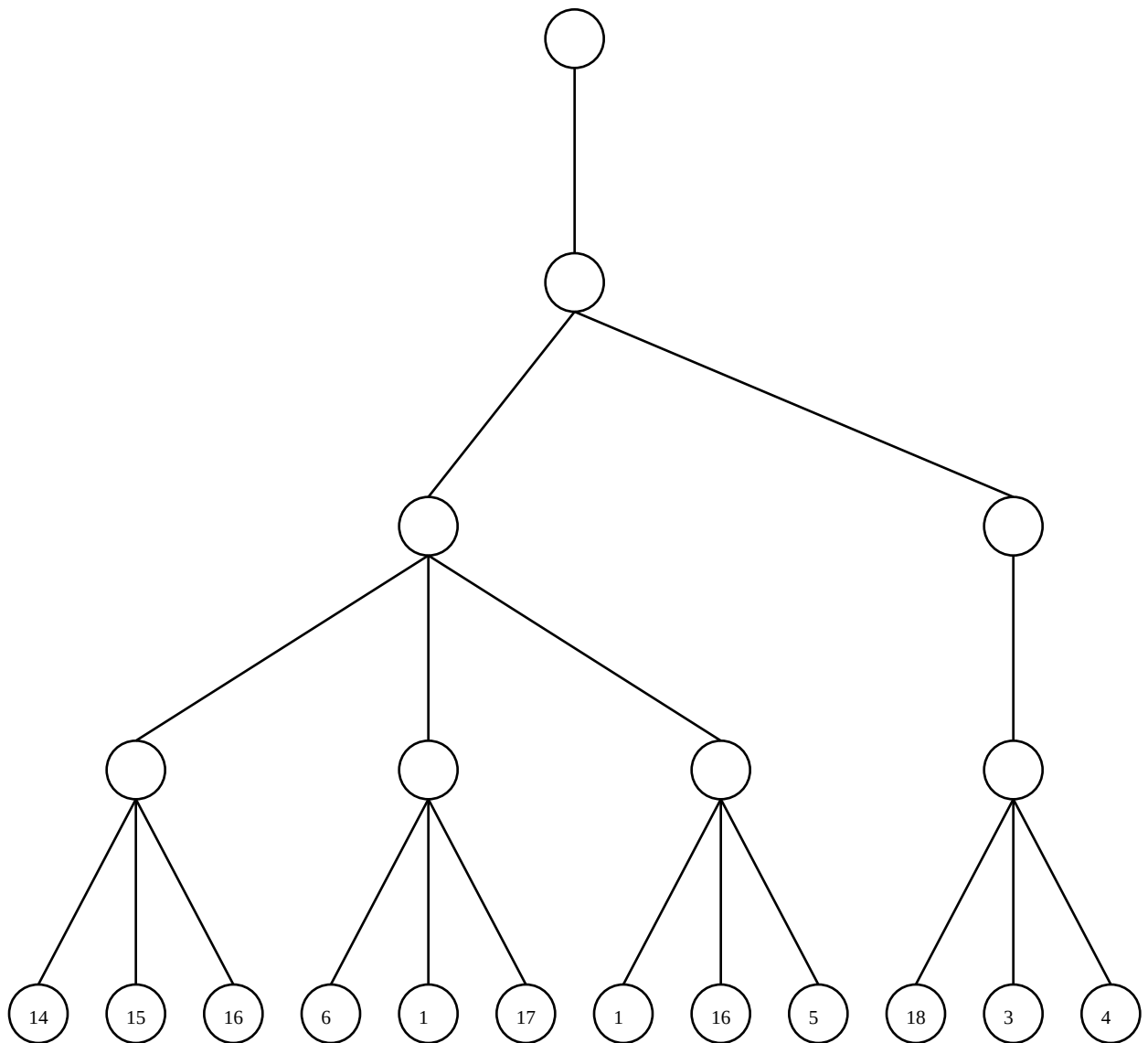
Question: For the given MinMax tree, what is the value at the root and how many leaf nodes are visited when applying MinMax with Alpha-Beta pruning?

Answer:

Root Value: 14

Leaves Visited with Alpha-Beta: 4

Instance 2:



Question: For the given MinMax tree, what is the value at the root and how many leaf nodes are visited when applying MinMax with Alpha-Beta pruning?

Answer:

Root Value: 3

Leaves Visited with Alpha-Beta: 8

Instance 3:

Leaves Visited with Alpha-Beta: 9